

ENS 492 – GRADUATION PROJECT

Multi Modal Fault Localization using an Autonomous Multi Agent System

Group Members: Lyan Alhavasli – 32714, Ahmet Furkan Ünal – 30680
, İbrahim Suat Gürsoy – 28974

Supervisor: Cemal Yilmaz

Outline

01 Background and Objectives

02 Project Overview and Setup

03 Methodology

04 Challenges

05 Results and Conclusion

06 Next Steps

Background: Debugging Crisis

✓ Software Complexity

Modern apps are becoming increasingly complex.

✓ Data Overload

Such apps have huge codebases and trace logs.

✓ Manual Effort

Root cause analysis requires hours of manual human inspection.

✓ Traditional Gap

Current tools identify *where* a bug is, but not *why* it happened and offer no explanations.

Project Overview

This is an AI-powered fault localization system for UI-driven applications (mobile/web) to determine which steps failed and the failure type, and implement a cascading failure analysis to produce a diagnostic report.

Core hypothesis: A Vision-Language Model (VLM) can observe what happened visually between two screenshots, and a separate reasoning LLM can then judge whether the application behaved correctly, producing a structured fault report.

Input

Recorded test execution:

- Prev/Post screenshots of every step
- JSON Files containing:
Action logs and
Runtime logs (console and network requests)
- 12 apps

Output

A human-readable fault report showing:

- Chains
- Confidence
- Suspicious step rankings

Motivation and Objectives

- **Beyond Simple Detection:** Multi-agent framework that identifies and explains failures with minimal human intervention.
- **Multi-modal Approach:** Framework that combines both vision and logic.
- **Developer as a Supervisor:** Transition, not replace, the role of the developer.



Autonomous Architecture: Multi-agent system using **LangGraph** to prevent the “anchoring bias”.



Multi-modal Analysis: Process screenshots, action logs, and network logs.



Root-cause Explanations: Human-readable reporting system.



Reliable Debugging: Reduce hallucinations and biased conclusions.

Project Setup



Orchestration

Built with **LangGraph** to create sequential graph, required for reproducible academic research



Model

Powered by **GPT-4o** for advanced visual observation and logical reasoning.



Validation

Used **Pydantic v2** to ensure all communication between the AI agents is strictly typed and reliable.



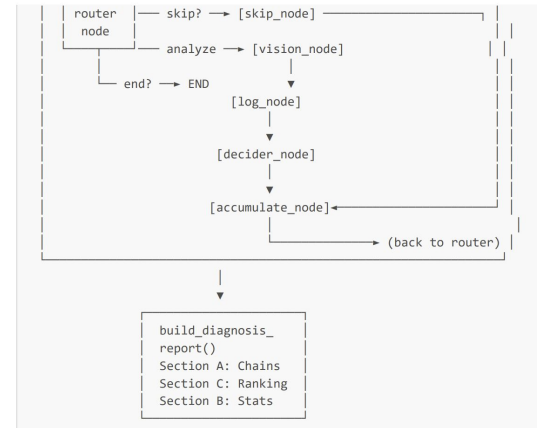
Methodology

System Architecture

Vision

Logic

Log





Challenges



Results and Conclusion





Next Steps

